

LEDGERIX PRO

Claude Project Brief

Context document for AI assistant — optimized for fast loading

Version 1.4 — May 17, 2026 — CONFIDENTIAL

PURPOSE OF THIS DOCUMENT

You are assisting Scott Hansbury, Founder & Business Strategist of Ledgerix Pro — an AI-powered autonomous bookkeeping firm deployed on Railway cloud. This brief gives you everything you need to assist without re-explanation. Read it fully before responding to any Ledgerix Pro question.

1. Who & What

Key	Value
Founder	Scott Hansbury — scott@ledgerixpro.com / admin@ledgerixpro.com
Company	Ledgerix Pro LLC — AI-powered bookkeeping firm
Location	Scottsdale / Phoenix, Arizona
Stage	Pre-beta — pipeline ready, no active beta clients pending Pattern B Full Phase 5
Codebase	/Users/scotthansbury/Projects/ledgerix-pro-core (local) github.com/LedgerixPro/ledgerix-pro-core (private)
Live URL	https://api.ledgerixpro.com
Dashboard	https://api.ledgerixpro.com/dashboard
Dashboard Secret	e877bb00219cd8758269d397743906d8995144ff7e4c2f9a69741574672e894c
Paperclip Company ID	f60117de-1131-433c-934f-3fe88bfaa163
Board API Key	pcp_board_railway_admin_key_2026
GHL Location ID	GhnRONQQVJiCKsdWoQFc
Railway DB	postgres://postgres:TsZlqUCeemXLEDyIHBwOcgmdEGSoTkKS@turntable.proxy.rlwy.net:32057/railway

1.1. Trust Tenet (established 2026-05-24)

No real clients — including Ledgerix Pro's own books — onboarded until the system is correct, trustworthy, and dialed in for security and safety of client funds. Applies uniformly. No partial-spec compliance on safety-critical write endpoints. Time is reference for planning, not a gate for go/no-go decisions.

Operating Principles (encoded in CLAUDE.md at repo root):

- Verify before assuming (grep callers, view imports, read authoritative docs before quoting)
- Session-end documentation discipline (EA + Brief + trackers + WIP docs always reflect committed reality)
- Locked decisions stay locked (no re-litigation across sessions)
- WIP docs (docs/wip/) are TRUTH for active multi-session work

WIP Convention: Multi-session architectural work has documents at docs/wip/<feature>.md capturing locked decisions, pending questions, rejected options, and append-only session logs. Read docs/wip/README.md for the convention.

2. Tech Stack at a Glance

Layer	Technology	Status
Agent Orchestration	Paperclip AI (claude_local adapter, claude-sonnet-4-6 model)	Live on Railway
Infrastructure	Railway cloud — auto-deploys on git push to master	Live
Database	External PostgreSQL on Railway	Live
CRM	GoHighLevel (GHL)	Live
Accounting	Xero (multi-tenant per-contact OAuth) + QBO (built, pending client)	Xero live
Language	TypeScript / Node.js (ESM, pnpm monorepo)	Live
DB Backups	Hourly automated .sql.gz snapshots in Railway container	Live
Client Portal	api.ledgerixpro.com/portal/{slug} or /{contactId}	Live
Xero Webhook	Registered at developer.xero.com — Invoices, Contacts, Credit Notes	Live, OK
A2P SMS	GHL Trust Center approval pending	Pending
Google Place ID	Claim Google Business profile first, then add to Railway env vars	Not yet set

⚠ RAILWAY CRITICAL

Never wrap Railway env var values in quotes. KEY=value not KEY="value". Quoted values broke ANTHROPIC_API_KEY auth. This has caused failures before.

3. Agent Roster (18 Active of 25 Configured)

Division	Agents	Key Schedules
Accounting (CFO)	Senior Bookkeeper, Ledger Specialist, Reconciliation Agent, AP Specialist, AR Specialist, Payroll, Reporter, Tax Liaison, Billing & Invoicing	Sentinel 6am daily; AP 6:30am daily; Tax 7am daily; Senior Bookkeeper digest Mon 8am
Operations (COO)	Sentinel, Onboarding, Quality Control, Audit & Compliance, Knowledge Base Manager	QC spot-check 7:30am Mon-Fri; QC weekly Fri 9am; Audit & Compliance Mon 10:15am
Revenue (CRO)	SDR (Laura), Client Success Manager, Referral & Reviews, Reactivation	Referral & Reviews Tue 9am; Reactivation 1st of month 9am
Dormant	Client Health Monitor (enable at 5 paying clients), Sales Outreach (50+ leads)	Client Health Monitor cron: 0 15 * * 1 (Mon 8am Arizona)

4. Build Status — All Horizons Complete

Horizon	Status	Summary
H1 Foundation	Done	GHL webhooks, Paperclip setup, dual-path auth, workspace registry
H2 Accounting Core	Done	Xero + QBO OAuth, Sentinel, Ledger Specialist, Reconciliation, Senior Bookkeeper, KB Manager
H3 Operations	Done	Dashboard, weekly digest, budget guardrails, diagnostic CRO funnel, nurture, SDR Laura, Reactivation
H4 Full Platform	Done	Client portal (slug URLs), 6 new agents, Railway migration, multi-tenant OAuth (H4-14), AR payment date intelligence
Phase 4 Accounting API	Phase 4 Accounting API	5 of 8 endpoints production-ready. Write endpoints (transactions/category, payments, invoices) deferred behind Phase 4c safety architecture.
Phase 4c Safety Architecture	In progress	4 of 5 pieces complete: pricing source of truth, threshold framework, customer dedupe with HITL, write-approval dispatcher (stub mode). Phase 4c.5 in progress — see docs/wip/phase-4c-5-write-endpoints-and-admin-api.md
H4 Deferred	Scale	Sales Outreach (50+ leads), Scale Pattern (50 clients), KB rules DB table

5. Critical IDs & References

Agent IDs (Key Agents)

Agent	Paperclip ID
Onboarding	2ae62352-b235-417f-ae97-99df11414ebb (maxTurnsPerRun: 40)
SDR (Laura)	2b4d27f8-5893-481c-a984-9203f4f1af8e
All others	Query: SELECT id, name FROM agents WHERE company_id = 'f60117de-...' ORDER BY name

Beta Clients

No active beta clients. Enyrgy Inc will be onboarded through the production pipeline once Pattern B Full Phase 5 ships.

GHL Custom Fields & Pipelines

Field / Pipeline	ID
service_tier field	Dh5rwdlahz6a37BAQDIIs
client_type field	Cf539co3LHJrm6wLAJQJ
ledgerix_workspace_id field	vmAT4OjG10QboXA2Jqjs (vestigial post-H4-14, retained for compat)
nurture_month field	sMQegZrU2giDsyaNKnjt
Sales Pipeline	dtgrQV0u9DB5EmxJGY9K
Clients Pipeline	EOq8U8BCqRMX9kM5g2qS
Churn Pipeline	A4SSmXmDnwPKGfxKcvut

Phase 4c Safety Architecture

Every write endpoint (POST /transactions/:txnId/category, POST /payments, POST /invoices) sits atop the Phase 4c safety layer. Endpoint flow: validate → safety checks → if pass: upstream write + audit + idempotency; if fail: create approval (202 Accepted) + audit + idempotency.

Components shipped (as of 2026-05-24):

- Pricing source of truth (4c.1)
— service_tier_pricing + client_pricing_overrides tables; getExpectedPriceCents(db, tier, isCharter, contactId?) service function. Effective-dating throughout.
- Threshold framework (4c.2)
— write_thresholds table; getApplicableThresholds, getMostSpecificThreshold, isThresholdExceeded service functions. Hierarchical: per-client overrides win over global defaults.

- **Customer dedupe with HITL (4c.3)** — `findOrCreateCustomer` returns `{customerId, action}` where action is one of 5 types (3 auto-proceed, 2 require approval). String-similarity utility (Levenshtein + name normalization).
- **Write-approval dispatcher (4c.4, stub mode)** — `services/accounting/write-approvals.ts`. 4 dot-namespaced approval types (`accounting.payment.threshold_exceeded`, `accounting.invoice.dedupe_ambiguous`, etc.) with typed payloads. Wired into `approvalService.approve()` for routing.

In progress: Phase 4c.5 — admin endpoints for safety-layer data management + re-implementing the three write endpoints atop the safety layer. See WIP doc for current state.

Architectural Decision Records:

- ADR-001: Pattern B Full API endpoints
- ADR-002: Phase 4b write endpoint design (idempotency, audit, two-phase failure)
- ADR-003: Phase 4c safety architecture (10 decisions, 3 amendments)

6. Service Tiers & Pricing

Updated May 17, 2026. Growth Engine Standard moved from \$499 to \$599 and Scale-Up repriced to \$999/\$1,299 to match market positioning for specialist trades and agency bookkeeping. First 10 clients receive Charter pricing.

Tier	Target	Charter	Standard	Key Features
The Foundation	Freelancers & Micro-business	\$199/mo	\$299/mo	Full bookkeeping, monthly P&L, tax deadline alerts
The Growth Engine	Local Trades & Service businesses	\$399/mo	\$599/mo	Foundation + job costing, ROC license tracking, TPT compliance
The Scale-Up	Agencies & Law Firms	\$999/mo	\$1,299/mo	Growth Engine + trust account compliance, E&O tracking, multi-entity

Setup & Migration Fee

All clients (including Charter) pay a one-time setup fee at onboarding. The fee covers chart of accounts review, vendor categorization rules, platform connection, workflow training, and (for prospects not currently on QBO or Xero) data migration. No waivers.

Tier	Setup Fee
The Foundation	\$249
The Growth Engine	\$349
The Scale-Up	\$1,200

The setup fee applies once at onboarding. It is non-refundable except per the 30-day satisfaction guarantee (see Strategic Plan).

Tier Qualifiers

Industry framing drives the marketing pitch; the qualifiers below drive the price quoted at the discovery call. Most qualifiers must match to land in a given tier; if a client splits across tiers (e.g. Growth Engine on volume but Scale-Up on entity count), price to the higher tier.

Qualifier	Foundation	Growth Engine	Scale-Up
Monthly transaction volume	Up to 75	75–300	300+
Bank/credit accounts	1–2	2–5	5+
Employees on payroll	0–2	3–15	10–25
Accounting integrations	1 platform	2–3 platforms (FSM, etc.)	3+ platforms
Annual revenue	Under \$500K	\$500K–\$3M	\$2M+
Job costing required	No	Yes	Yes
Trust/multi-entity	No	No	Yes
Industry flags	None	ROC, TPT, seasonal	IOLTA, E&O, accrual

CHARTER PRICING WINDOW

Charter pricing applies to the first 10 paying clients only. After client #10, the Charter column closes and new clients onboard at Standard pricing.

Charter benefit follows the client across tiers. As long as service is continuous, Charter clients keep Charter pricing for the duration of their relationship with Ledgerix Pro — even as their business grows from one tier to the next. A Foundation Charter client whose qualifiers move them to Growth Engine pays Growth Engine Charter (\$399/mo), not Standard (\$599/mo). Same applies for Scale-Up: Charter clients who grow into Scale-Up pay Scale-Up Charter (\$999/mo), not Standard (\$1,299/mo). The benefit also applies on tier downgrades: a Scale-Up Charter client whose qualifiers shrink to Growth Engine pays Growth Engine Charter, not Standard.

Continuity rule. If a Charter client cancels service and later returns, they re-enter at Standard pricing for whichever tier they qualify for. The Charter benefit is granted to the first 10 paying clients as a thank-you for choosing Ledgerix Pro early; it is not transferable, sellable, or recoverable after cancellation.

7. Architecture Patterns to Know

Multi-Tenant OAuth (H4-14)

- Each client has their own Xero/QBO connection keyed by GHL contact_id in accounting_connections table
- Connect URL:
https://api.ledgerixpro.com/api/oauth/xero/connect?contactId={ghlContactId}
- All agents call: getNewTransactions(db, COMPANY_ID, contact.id, sinceDate)
- Unique constraint: (company_id, platform, contact_id) with NULLS NOT DISTINCT

Slug-Based Client Portal

- https://api.ledgerixpro.com/portal/{company-slug} → 302 redirect → /portal/{contactId}
- Slug = company name lowercased, spaces → hyphens, special chars stripped. Automatic for all clients.

Escalation Chain (All Agents)

Senior Bookkeeper → CFO → Scott

No agent contacts Scott directly for routine issues. Scott only for Level 3 events (fraud, breach, cancellation).

Webhook & Agent Trigger Flow

- GHL fires webhook → /api/webhooks/ghl → Dispatcher → Paperclip creates issue → Agent wakes up
- Key events: contact.created, contact.updated, contact.sdr_ready, contact.replied
- Agents discriminate mode by issue title (e.g. 'Weekly QC Review' vs daily, 'Monthly Payroll Review' vs bi-weekly)

8. Scale Triggers

Trigger	Action
5 paying clients	Enable Client Health Monitor: cron 0 15 * * 1 (Monday 8am Arizona / 15:00 UTC). Register at /companies/f60117de.../routines
10 clients	Charter pricing window closes — new clients move to Standard pricing
Each 25-client milestone	Revisit agent budget caps (~\$1,225 at 25, ~\$3,110 at 75, ~\$5,480 at 150, ~\$9,700 at 300)
~50 leads	Enable Sales Outreach agent (H4-9)
~50 clients	Scale Pattern webhooks + KB rules DB table (H4-12/13)

9. Database Migrations

Current migration head: 0067_last_gateway.sql

Recent migrations relevant to Phase 4 work:

- 0064 — activity_log.status column + idempotency_keys table (Phase 4b foundation)
- 0065 — service_tier_pricing + client_pricing_overrides (Phase 4c.1)
- 0066 — write_thresholds (Phase 4c.2)
- 0067 — activity_log.company_id nullable for system-scoped admin operations (Phase 4c.5 Decision B)

10. Test Status

Current targeted test baseline: 145 tests passing

- 62 accounting routes
- 14 idempotency
- 10 pricing (4c.1)
- 14 thresholds (4c.2)
- 25 string-similarity (4c.3)
- 9 findOrCreateCustomer (4c.3)
- 11 write-approvals (4c.4)

Note on full suite: Running `pnpm exec vitest run` (entire server, 1240 tests) has known flakiness in unrelated `workspace-runtime.test.ts` due to parallel-run resource contention. Same tests pass cleanly when run in isolation. Pre-existing infrastructure issue unrelated to Phase 4c work. Verified 2026-05-24.

11. Open / Pending Items

- A2P SMS — GHL Trust Center approval pending (Laura's SMS outreach queued, email works fine)
- Google Business Profile — claim at business.google.com, then add `GOOGLE_PLACE_ID` to Railway env vars
- Client portal auth — add proper token auth when first paying client requests portal access
- Xero bank transactions — NOT available as webhook events (Xero limitation). Sentinel daily 6am poll handles these.

12. How to Help Scott Effectively

- Scott uses Claude Code for all code changes — give complete, ready-to-paste instructions
- Always check Railway env var syntax: no quotes around values (KEY=value not KEY="value")
- RESET.md in repo root is the authoritative recovery runbook — reference it for infrastructure questions
- AGENTS.md files in agents/{name}/ are authoritative for agent behavior — suggest edits there for behavioral changes
- When creating Paperclip issues via API: POST to /api/companies/{companyId}/issues with assigneeAgentId and status: in_progress
- Onboarding agent has maxTurnsPerRun: 40 in DB — if it fails with error_max_turns, UPDATE in DB, not code
- Scott's background: CFO at multiple companies, CEO of Enyrgy, VP Marketing/Biz Dev — speak at executive + technical level
- All 17 cron routines are live and firing. 18 active agents. H1-H4 complete.

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Scott Hansbury | Founder & Business Strategist | Version 1.2 | May 17, 2026